oblig 2

$f(x) \quad e^{-\frac{x}{4}} \cdot \tan^{-1}(x)$

$u = e^{-\frac{x}{4}}$

$u' = -\frac{1}{4} \cdot e^{-\frac{x}{4}}$

$v = \tan^{-1}(x)$

$v' = \frac{1}{1+x^2}$

$f'(x) = -\frac{1}{4} e^{-\frac{x}{4}} \cdot \tan^{-1}(x) + e^{-\frac{x}{4}} \cdot \frac{1}{1+x^2}$

$e^{-\frac{x}{4}} \left(-\frac{1}{4} \cdot \tan^{-1}(x) + \frac{1}{1+x^2} \right)$

$\left(-\frac{\tan^{-1}(x)}{4} + \frac{1}{1+x^2} \right) \cdot (-1)$

$\underline{\tan^{-1}(x) - \frac{4}{1+x^2}}$

$f'(2) > 0 \quad , \quad f'(1) < 0$

midtpunkt $= 1{,}5 \quad x = -0{,}2479$

$x = 1{,}6 \quad = -0{,}1113$

$x = 1{,}65 = -0{,}0486$

$x = 1{,}69 = -8{,}27 \cdot 10^{-4}$

$\underline{x = 1{,}6907} = -9{,}522 \cdot 10^{-6}$

$= 0{,}00000 9522$

<u>toppunkt når $x = 1{,}6907$</u>